Data Analytics Mini Project

Data Set Link: Student Lifestyle and Academic Performance Dataset

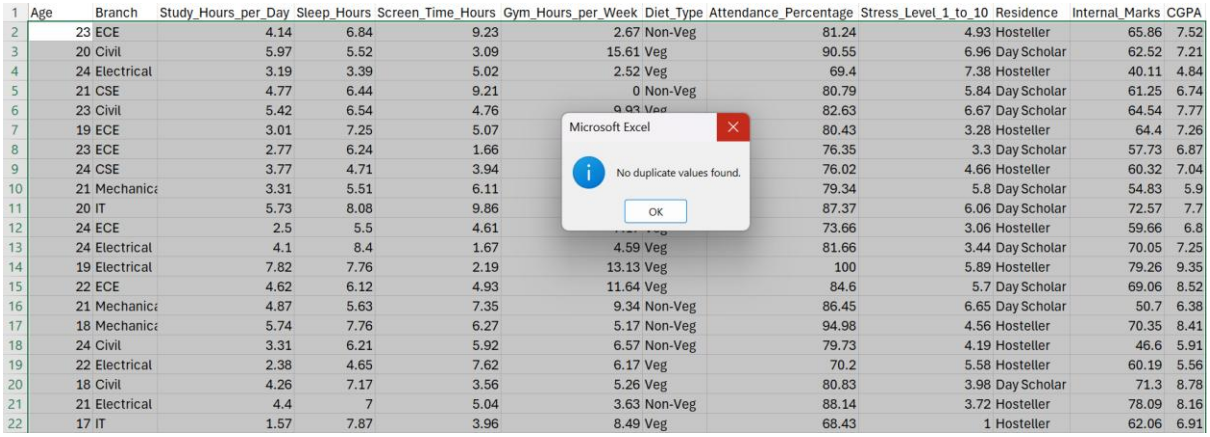
Website: www.Kaggle.com

Objective

The objective of this project is to analyze student lifestyle and academic performance data, perform data cleaning and transformation using Excel, and develop an interactive dashboard in Power BI to generate meaningful insights.

1. Data Pre-processing using Excel

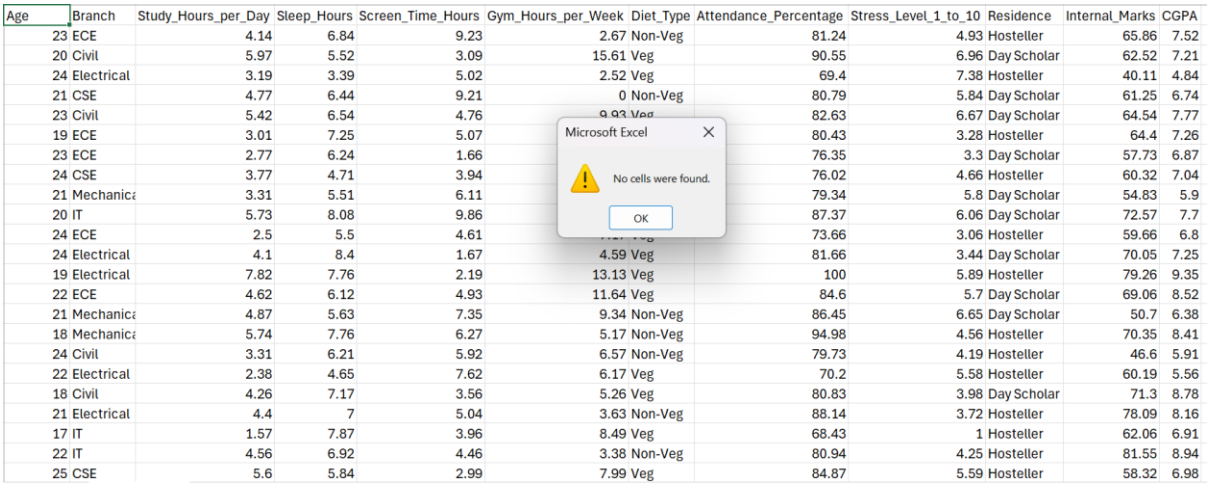
Step1: Removing duplicates



Age	Branch	Study_Hours_per_Day	Sleep_Hours	Screen_Time_Hours	Gym_Hours_per_Week	Diet_Type	Attendance_Percentage	Stress_Level_1_to_10	Residence	Internal_Marks	CGPA
23	ECE	4.14	6.84	9.23	2.67	Non-Veg	81.24	4.93	Hosteller	65.86	7.52
20	Civil	5.97	5.52	3.09	15.61	Veg	90.55	6.96	Day Scholar	62.52	7.21
24	Electrical	3.19	3.39	5.02	2.52	Veg	69.4	7.38	Hosteller	40.11	4.84
21	CSE	4.77	6.44	9.21	0	Non-Veg	80.79	5.84	Day Scholar	61.25	6.74
23	Civil	5.42	6.54	4.76	9.93	Veg	82.63	6.67	Day Scholar	64.54	7.77
19	ECE	3.01	7.25	5.07	[illegible]	[illegible]	80.43	3.28	Hosteller	64.4	7.26
23	ECE	2.77	6.24	1.66	[illegible]	[illegible]	76.35	3.3	Day Scholar	57.73	6.87
24	CSE	3.77	4.71	3.94	[illegible]	[illegible]	76.02	4.66	Hosteller	60.32	7.04
21	Mechanica	3.31	5.51	6.11	[illegible]	[illegible]	79.34	5.8	Day Scholar	54.83	5.9
20	IT	5.73	8.08	9.86	[illegible]	[illegible]	87.37	6.06	Day Scholar	72.57	7.7
24	ECE	2.5	5.5	4.61	[illegible]	[illegible]	73.66	3.06	Hosteller	59.66	6.8
24	Electrical	4.1	8.4	1.67	4.59	Veg	81.66	3.44	Day Scholar	70.05	7.25
19	Electrical	7.82	7.76	2.19	13.13	Veg	100	5.89	Hosteller	79.26	9.35
22	ECE	4.62	6.12	4.93	11.64	Veg	84.6	5.7	Day Scholar	69.06	8.52
21	Mechanica	4.87	5.63	7.35	9.34	Non-Veg	86.45	6.65	Day Scholar	50.7	6.38
18	Mechanica	5.74	7.76	6.27	5.17	Non-Veg	94.98	4.56	Hosteller	70.35	8.41
24	Civil	3.31	6.21	5.92	6.57	Non-Veg	79.73	4.19	Hosteller	46.6	5.91
22	Electrical	2.38	4.65	7.62	6.17	Veg	70.2	5.58	Hosteller	60.19	5.56
18	Civil	4.26	7.17	3.56	5.26	Veg	80.83	3.98	Day Scholar	71.3	8.78
21	Electrical	4.4	7	5.04	3.63	Non-Veg	88.14	3.72	Hosteller	78.09	8.16
17	IT	1.57	7.87	3.96	8.49	Veg	68.43	1	Hosteller	62.06	6.91

The dataset was checked for duplicate records using Excel's Remove Duplicates feature. No duplicate records were found.

Step2: Handling missing values



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21	CSE	4.77	6.44	9.21	0	Non-Veg	80.79	5.84	Day Scholar	61.25	6.74
23	Civil	5.42	6.54	4.76	9.93	Veg	82.63	6.67	Day Scholar	64.54	7.77
19	ECE	3.01	7.25	5.07	[illegible]	[illegible]	80.43	3.28	Hosteller	64.4	7.26
23	ECE	2.77	6.24	1.66	[illegible]	[illegible]	76.35	3.3	Day Scholar	57.73	6.87
24	CSE	3.77	4.71	3.94	[illegible]	[illegible]	76.02	4.66	Hosteller	60.32	7.04
21	Mechanica	3.31	5.51	6.11	[illegible]	[illegible]	79.34	5.8	Day Scholar	54.83	5.9
20	IT	5.73	8.08	9.86	[illegible]	[illegible]	87.37	6.06	Day Scholar	72.57	7.7
24	ECE	2.5	5.5	4.61	[illegible]	[illegible]	73.66	3.06	Hosteller	59.66	6.8
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21	Mechanica	4.87	5.63	7.35	9.34	Non-Veg	86.45	6.65	Day Scholar	50.7	6.38
18	Mechanica	5.74	7.76	6.27	5.17	Non-Veg	94.98	4.56	Hosteller	70.35	8.41
24	Civil	3.31	6.21	5.92	6.57	Non-Veg	79.73	4.19	Hosteller	46.6	5.91
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21	Electrical	4.4	7	5.04	3.63	Non-Veg	88.14	3.72	Hosteller	78.09	8.16
17	IT	1.57	7.87	3.96	8.49	Veg	68.43	1	Hosteller	62.06	6.91
22	IT	4.56	6.92	4.46	3.38	Non-Veg	80.94	4.25	Hosteller	81.55	8.94
25	CSE	5.6	5.84	2.99	7.99	Veg	84.87	5.59	Hosteller	58.32	6.98

The dataset was examined for missing values using Excel's Go To Special (Blanks) feature and filter validation. No missing values were detected across the dataset. Therefore, no imputation or removal was necessary

Step3: Standardizing Formats

Age	Branch	Study_Hou	Sleep_Hours	Screen_Time	Gym_Hours	Diet_Type	Attendance	Stress_Level	Resider	Internal_Marks	CGPA
23	ECE	4.14	6.84	9.23	2.67	Non-Veg	81.24	4.93	Hosteller	65.86	7.52
20	Civil	5.97	5.52	3.09	15.61	Veg	90.55	6.96	Day Schola	62.52	7.21
24	Electrical	3.19	3.39	5.02	2.52	Veg	69.40	7.38	Hosteller	40.11	4.84
21	CSE	4.77	6.44	9.21	0	Non-Veg	80.79	5.84	Day Schola	61.25	6.74
23	Civil	5.42	6.54	4.76	9.93	Veg	82.63	6.67	Day Schola	64.54	7.77
19	ECE	3.01	7.25	5.07	13.89	Non-Veg	80.43	3.28	Hosteller	64.40	7.26
23	ECE	2.77	6.24	1.66	8.00	Non-Veg	76.35	3.30	Day Schola	57.73	6.87
24	CSE	3.77	4.71	3.94	9.44	Veg	76.02	4.66	Hosteller	60.32	7.04
21	Mechanical	3.31	5.51	6.11	17.94	Veg	79.34	5.80	Day Schola	54.83	5.90
20	IT	5.73	8.08	9.86	13.97	Non-Veg	87.37	6.06	Day Schola	72.57	7.70
24	ECE	2.50	5.50	4.61	7.17	Veg	73.66	3.06	Hosteller	59.66	6.80
24	Electrical	4.10	8.40	1.67	4.59	Veg	81.66	3.44	Day Schola	70.05	7.25
19	Electrical	7.82	7.76	2.19	13.13	Veg	100.00	5.89	Hosteller	79.26	9.35
22	ECE	4.62	6.12	4.93	11.64	Veg	84.60	5.70	Day Schola	69.06	8.52
21	Mechanical	4.87	5.63	7.35	9.34	Non-Veg	86.45	6.65	Day Schola	50.70	6.38
18	Mechanical	5.74	7.76	6.27	5.17	Non-Veg	94.98	4.56	Hosteller	70.35	8.41
24	Civil	3.31	6.21	5.92	6.57	Non-Veg	79.73	4.19	Hosteller	46.60	5.91
22	Electrical	2.38	4.65	7.62	6.17	Veg	70.20	5.58	Hosteller	60.19	5.56
18	Civil	4.26	7.17	3.56	5.26	Veg	80.83	3.98	Day Schola	71.30	8.78
21	Electrical	4.40	7.00	5.04	3.63	Non-Veg	88.14	3.72	Hosteller	78.09	8.16
17	IT	1.57	7.87	3.96	8.49	Veg	68.43	1.00	Hosteller	62.06	6.91
22	IT	4.56	6.92	4.46	3.38	Non-Veg	80.94	4.25	Hosteller	81.55	8.94
25	CSE	5.60	5.84	2.99	7.99	Veg	84.87	5.59	Hosteller	58.32	6.98

The dataset was standardized by correcting column headers and ensuring consistent formatting across all fields. Numeric columns were formatted with appropriate decimal precision, and categorical variables were verified for consistent spelling and capitalization using filter validation.

Step4: Creating calculated fields

Sleep_Hours	Screen_Time	Gym_Hours	Health_Index	Diet_Type	Attendance	Stress_Level	Stress_Category	Residence	Internal_Marks	CGPA	Performance_Level
3.51	4.84	5.11	8.62	Non-Veg	63.67	7.14	High	Hosteller	40.84	4.00	Needs Improvement
4.33	1.83	5.01	9.34	Veg	71.84	3.69	Low	Hosteller	43.35	4.00	Needs Improvement
4.15	2.05	0.00	4.15	Veg	66.41	5.60	Moderate	Day Scholar	35.00	4.09	Needs Improvement
3.00	5.82	2.16	5.16	Veg	73.95	6.62	Moderate	Day Scholar	46.26	4.10	Needs Improvement
4.52	6.10	8.62	13.14	Veg	74.50	8.00	High	Day Scholar	40.08	4.21	Needs Improvement
5.94	6.57	0.00	5.94	Veg	70.63	3.48	Low	Hosteller	40.97	4.45	Needs Improvement
3.19	2.87	7.76	10.95	Non-Veg	69.31	7.89	High	Day Scholar	44.91	4.65	Needs Improvement
3.46	5.68	10.14	13.60	Veg	62.19	3.75	Low	Hosteller	39.71	4.65	Needs Improvement
3.39	5.02	2.52	5.91	Veg	69.40	7.38	High	Hosteller	40.11	4.84	Needs Improvement
4.11	2.81	12.80	16.91	Veg	74.78	9.74	High	Day Scholar	36.22	5.03	Needs Improvement
4.51	5.73	2.89	7.40	Veg	61.36	1.38	Low	Hosteller	40.90	5.14	Needs Improvement
3.00	5.40	6.98	9.98	Veg	88.11	8.74	High	Day Scholar	42.30	5.21	Needs Improvement
6.07	3.42	11.27	17.34	Non-Veg	77.74	4.14	Moderate	Hosteller	51.47	5.22	Needs Improvement
3.85	5.22	10.80	14.65	Veg	79.41	8.70	High	Hosteller	43.19	5.28	Needs Improvement
4.67	3.76	12.49	17.16	Veg	61.91	1.83	Low	Day Scholar	53.47	5.29	Needs Improvement
4.18	5.89	15.94	20.12	Non-Veg	82.36	7.49	High	Day Scholar	53.69	5.33	Needs Improvement
5.90	6.12	11.64	17.54	Non-Veg	80.02	5.41	Moderate	Hosteller	50.61	5.38	Needs Improvement
4.79	6.73	0.43	5.22	Non-Veg	64.75	5.54	Moderate	Day Scholar	45.48	5.40	Needs Improvement
5.29	1.53	8.83	14.12	Veg	63.27	3.17	Low	Day Scholar	39.94	5.42	Needs Improvement
4.86	6.33	6.84	11.70	Non-Veg	71.80	6.05	Moderate	Hosteller	50.42	5.45	Needs Improvement
5.37	5.76	0.00	5.37	Veg	74.63	3.32	Low	Day Scholar	43.38	5.50	Needs Improvement
5.30	7.45	2.77	8.07	Veg	87.39	7.36	High	Day Scholar	45.77	5.50	Needs Improvement
4.91	5.81	10.32	15.23	Veg	92.50	9.66	High	Day Scholar	48.89	5.51	Needs Improvement

Created New colums as

Performance Level: Categorized students as Excellent, Average, or Needs Improvement based on CGPA ranges using IF logic.

Stress Category: Grouped stress scores into Low, Moderate, and High levels using conditional IF formulas.

Health Index: Calculated by summing Sleep Hours and Gym Hours to reflect wellness balance.

Step5: Filtering and sorting

Sleep_Hours	Screen_Time	Gym_Hours	Health_Index	Diet_Type	Attendance	Stress_Level	Stress_Category	Residence	Internal_Marks	CGPA	Performance_Level
9.26	7.19	6.05	15.31	Veg	88.30	4.66	Moderate	Hosteller	90.79	10.00	Excellent
7.92	2.21	18.13	26.05	Non-Veg	95.26	5.40	Moderate	Hosteller	90.39	10.00	Excellent
9.13	7.92	7.85	16.98	Veg	87.13	2.53	Low	Hosteller	85.80	10.00	Excellent
8.45	3.84	12.67	21.12	Veg	92.57	3.53	Low	Day Scholar	77.75	10.00	Excellent
8.34	3.28	3.59	11.93	Veg	76.75	1.00	Low	Day Scholar	84.53	10.00	Excellent
7.34	6.46	0.00	7.34	Non-Veg	100.00	7.55	High	Day Scholar	85.61	9.94	Excellent
8.07	2.98	12.96	21.03	Veg	82.76	3.05	Low	Hosteller	86.69	9.92	Excellent
8.57	3.96	8.24	16.81	Non-Veg	95.88	4.56	Moderate	Hosteller	89.41	9.88	Excellent
9.22	7.14	5.21	14.43	Non-Veg	74.77	1.79	Low	Day Scholar	93.93	9.69	Excellent
8.25	4.95	0.00	8.25	Veg	88.87	2.30	Low	Hosteller	85.85	9.61	Excellent
7.66	5.62	0.21	7.87	Non-Veg	87.03	2.03	Low	Hosteller	87.09	9.57	Excellent
6.21	4.73	0.00	6.21	Veg	94.68	8.38	High	Hosteller	84.36	9.47	Excellent
6.13	2.12	9.23	15.36	Veg	90.44	4.86	Moderate	Day Scholar	77.42	9.39	Excellent
9.41	4.70	7.50	16.91	Veg	87.76	1.91	Low	Hosteller	77.77	9.38	Excellent
6.77	7.07	1.80	8.57	Veg	98.26	6.48	Moderate	Day Scholar	85.31	9.37	Excellent
7.76	2.19	13.13	20.89	Veg	100.00	5.89	Moderate	Hosteller	79.26	9.35	Excellent
8.62	6.68	3.02	11.64	Veg	76.36	1.19	Low	Hosteller	85.46	9.34	Excellent
8.25	3.52	0.00	8.25	Veg	91.65	2.72	Low	Day Scholar	78.13	9.24	Excellent
8.76	2.62	11.52	20.28	Veg	92.34	3.58	Low	Day Scholar	79.17	9.16	Excellent
8.36	3.71	7.73	16.09	Non-Veg	85.51	2.69	Low	Day Scholar	79.38	9.16	Excellent
8.11	2.23	8.91	17.02	Veg	83.27	2.38	Low	Day Scholar	77.07	9.16	Excellent
7.38	4.49	4.12	11.50	Non-Veg	89.10	8.38	High	Day Scholar	88.83	9.15	Excellent
7.33	7.63	13.67	21.00	Non-Veg	89.31	5.22	Moderate	Hosteller	81.38	9.15	Excellent

Conditional filters were applied to the calculated fields to isolate specific student groups. For example, filters were used to view students with *High Stress* and *Low Health Index*, or those with *Excellent Performance Level* and *Hosteller* residence type. Sorting was applied to rank students by CGPA and Health Index, enabling focused analysis of academic and wellness patterns.

Step6: Creating Pivot Tables

Row Labels	Average of CGPA
Civil	7.53
CSE	7.20
ECE	7.30
Electrical	7.23
IT	7.31
Mechanical	7.32
Grand Total	**7.32**

Row Labels	Average of Attendance
Day Scholar	79.57
Hosteller	80.59
Grand Total	**80.08**

Row Labels	Count of Performance_Level
Average	294
Excellent	245
Good	386
Needs Improvement	75
Grand Total	**1000**

Branch

- Civil
- CSE
- ECE
- Electrical
- IT
- Mechanical

Residence

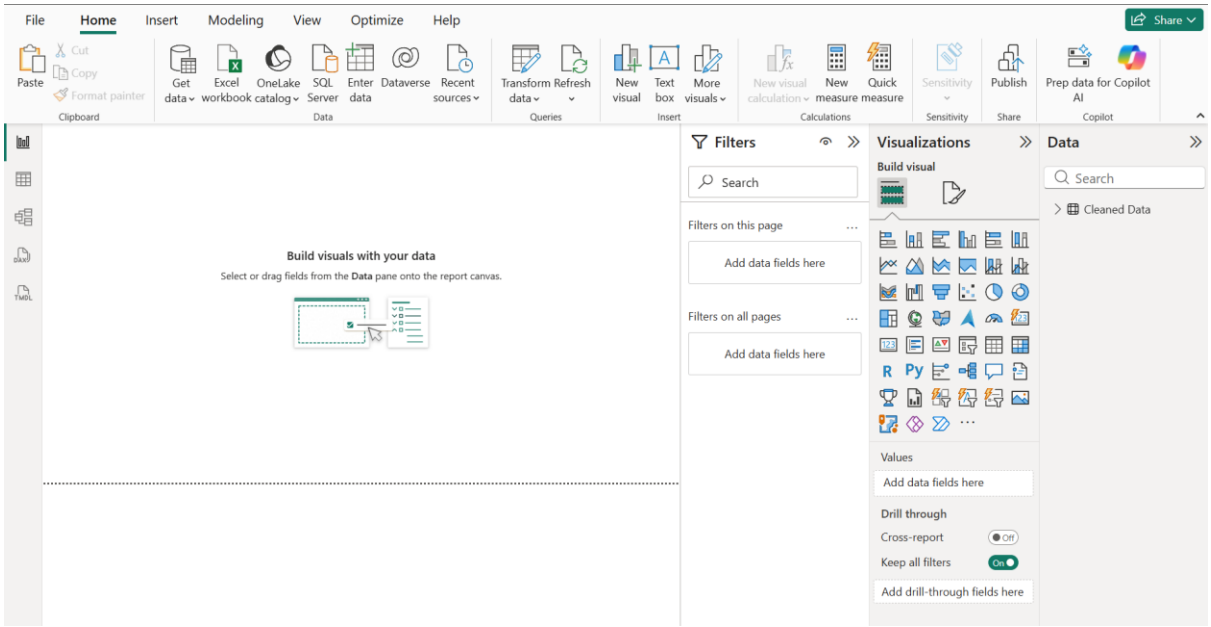
- Day Scholar
- Hosteller

Pivot tables were created to summarize and analyze student performance data. The average CGPA was calculated across branches, average attendance was compared by residence type, and student distribution was analyzed based on performance levels

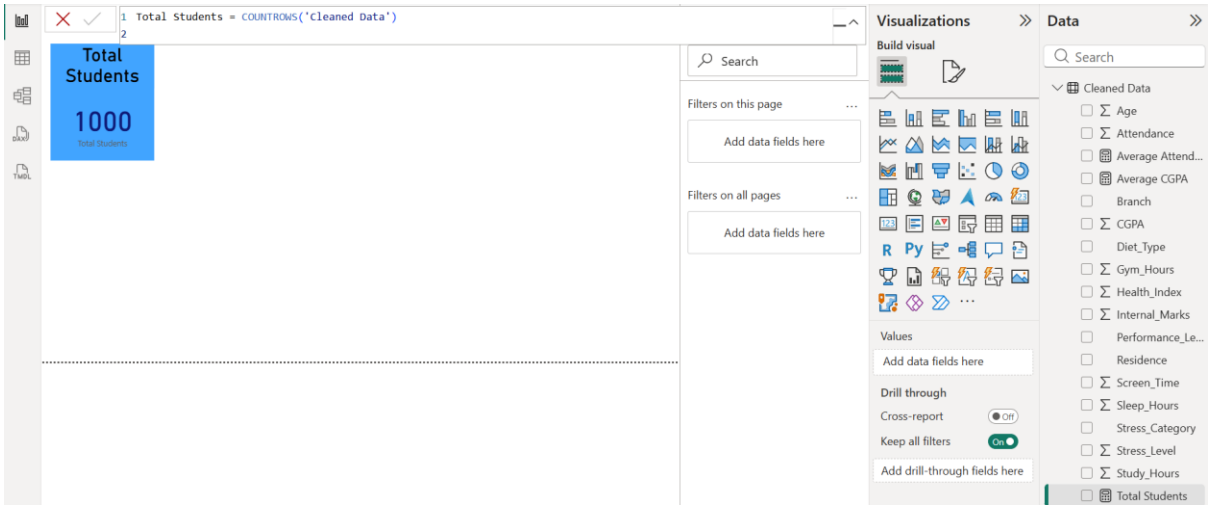
Slicers were added to enable interactive filtering by Branch and Residence, allowing dynamic exploration of performance trends.

2. Data Visualization using Power BI

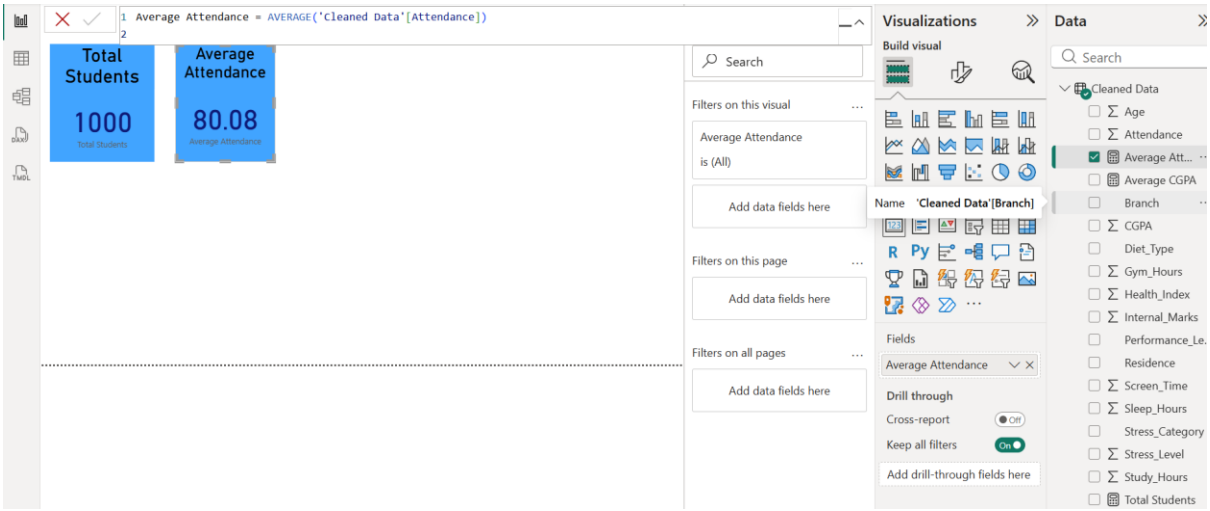
Step1: Import Cleaned Data into Power Bi



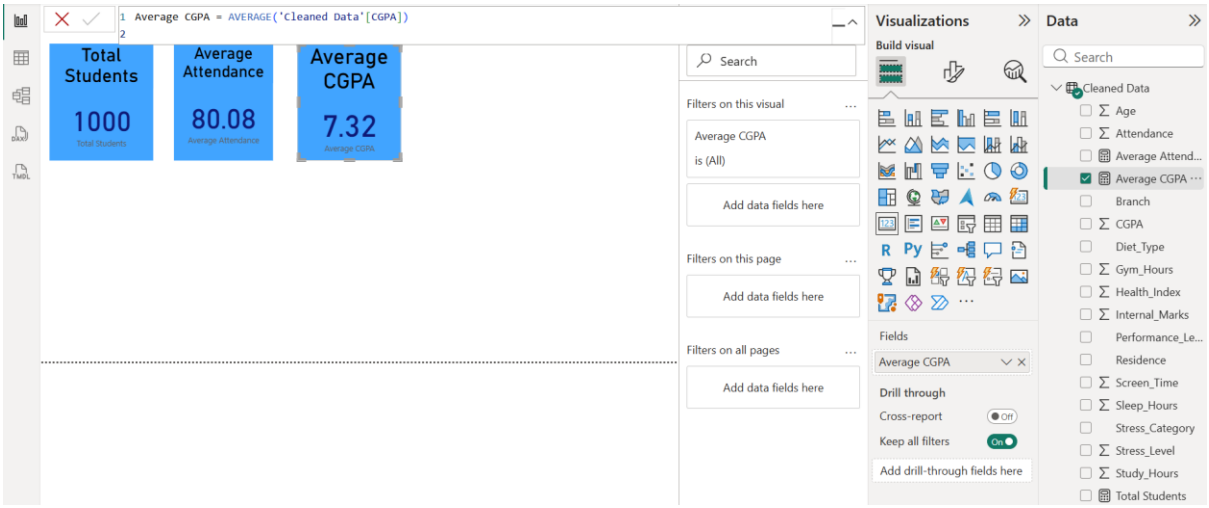
Step2: Created a DAX measure to calculate Total Students using COUNTROWS function.



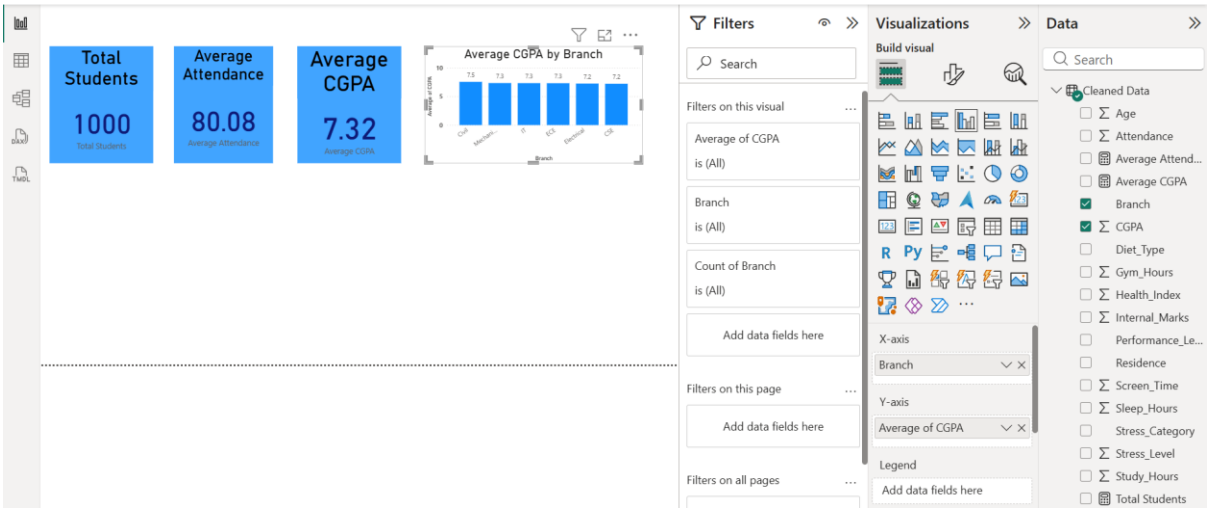
Step3: created a DAX measure to find the total number of students Average of attendance



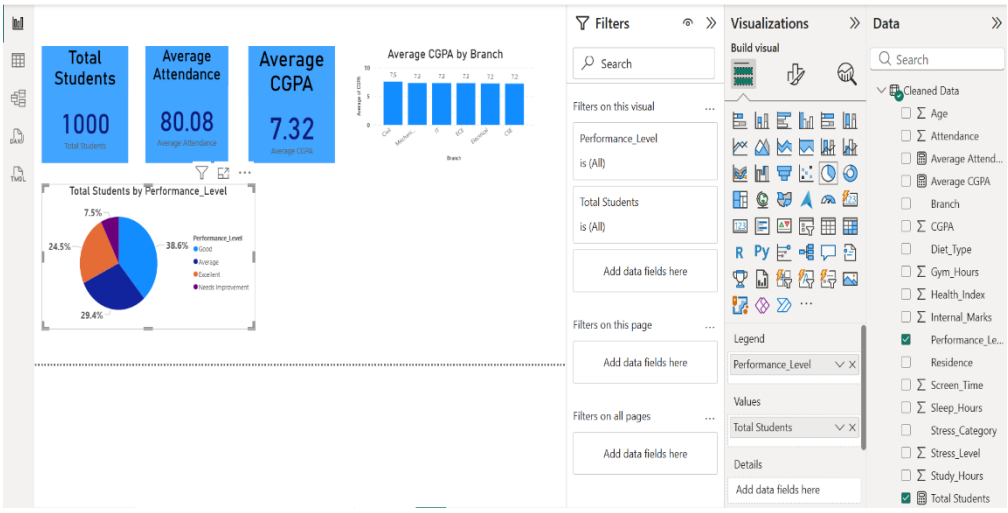
Step4: created a DAX measure to find the total students Average of CGPA



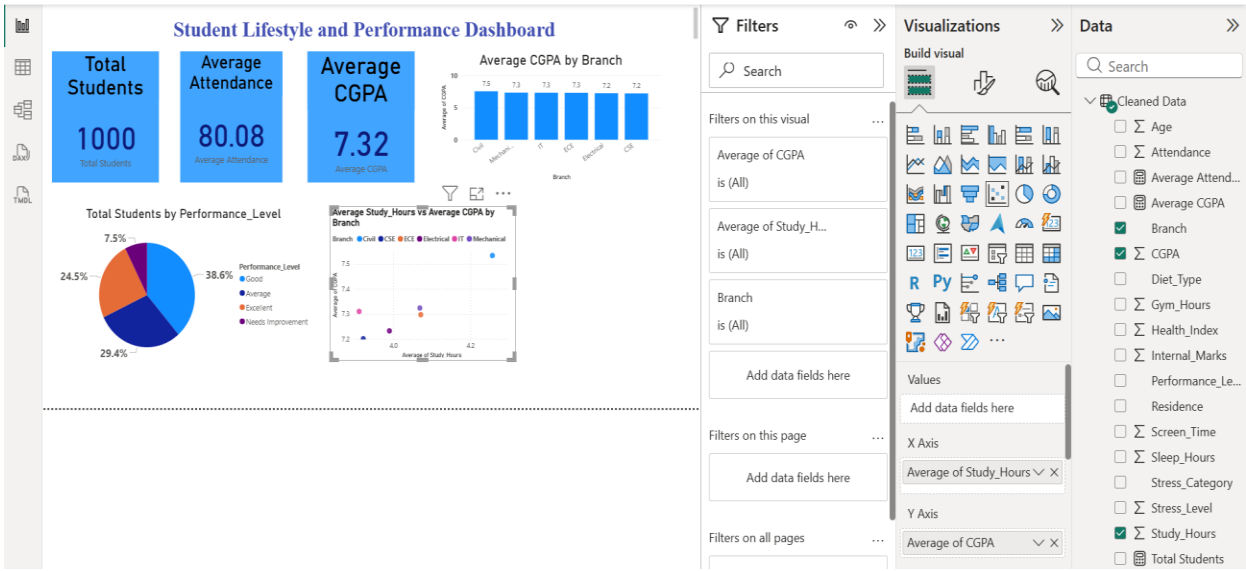
Step5: Total Students Average of CGPA by Branch Category using Clustered Column Chart



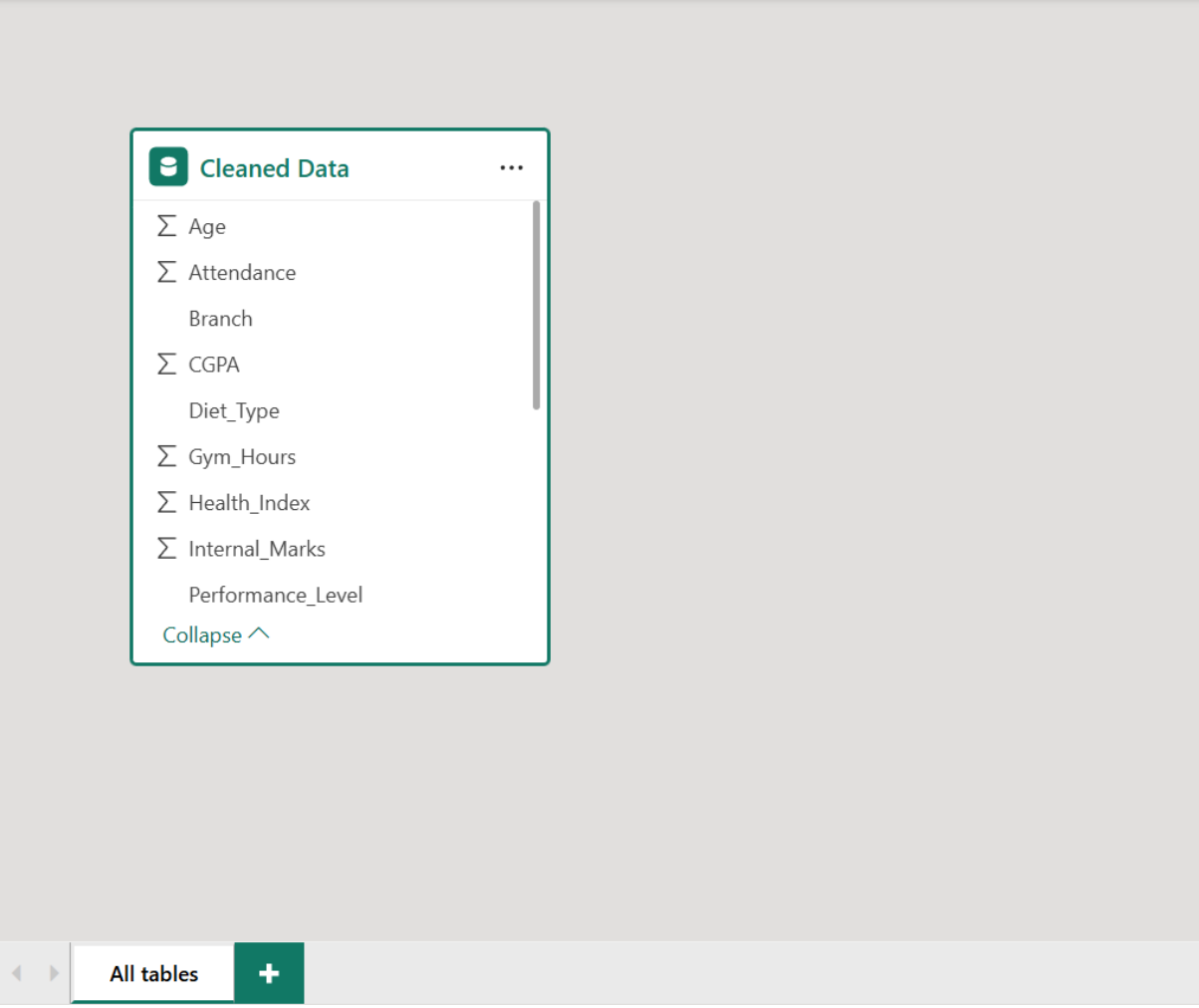
Step6: Total Students by Performance Level created using Pie Chart



Step7: A scatter plot was used to analyze the relationship between study hours and CGPA

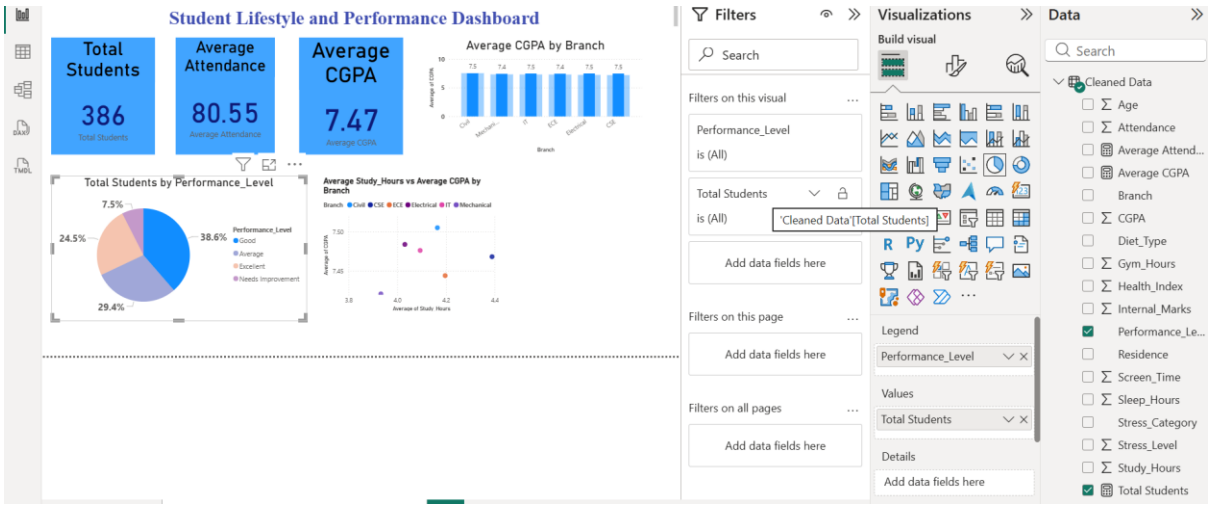


Step8: Data Modelling

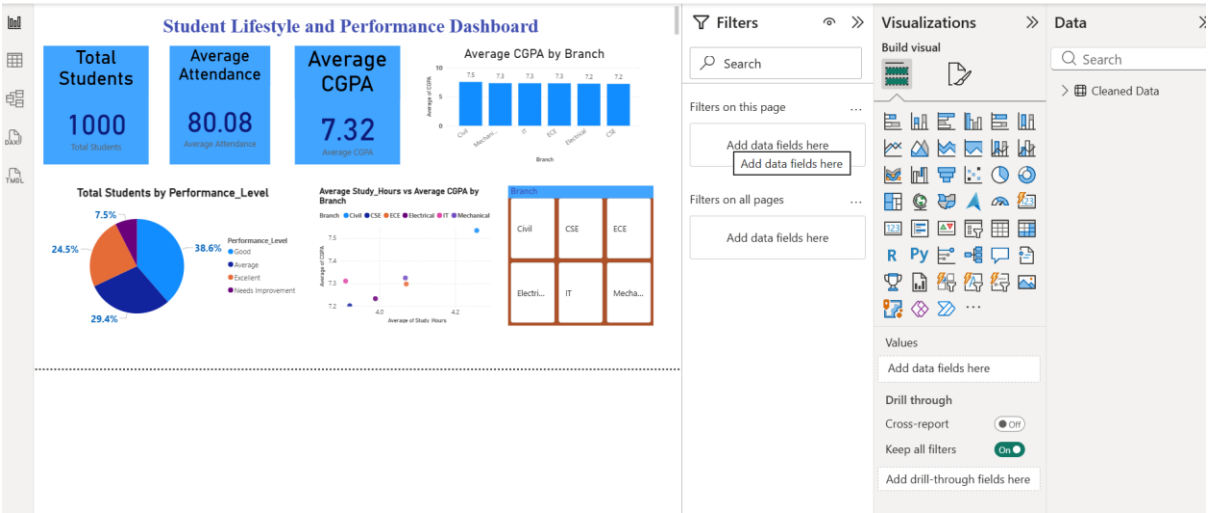


Since the dataset contains only one table, no relationships were required. The model view shows the structured table used for analysis.

Step9: Multiple visualizations that communicates insights clearly were created including clustered column chart, pie chart, scatter plot, slicers, and card visuals. The dashboard was formatted with proper titles, legends, labels, and interactive filtering to enhance clarity and usability.



Final Dashboard with Slicers



The final Power BI dashboard presents a comprehensive analysis of student lifestyle and academic performance. It includes key performance indicators such as Total Students, Average Attendance, and Average CGPA. A clustered column chart compares average CGPA across different branches, while a pie chart shows the distribution of students based on performance level. The scatter plot visualizes the relationship between average study hours and average CGPA across branches. A slicer is included to enable interactive filtering by branch, allowing dynamic analysis of all visualizations.

Conclusion

This project successfully analyzed student lifestyle and academic performance using Excel for preprocessing and Power BI for interactive visualization. The dashboard provides insights into branch-wise performance, attendance trends, stress distribution, and study behavior correlation. The interactive design allows dynamic filtering and deeper exploration of academic patterns.